

This review will explore some causes and attempted solutions of diminished workplace productivity. This section will focus on the effects of context switching, interruptions, and team size. In 2009, Leroy et al. introduced their idea of *attention residue*, an effect where a worker displays reduced focus after task switching when they don't receive 'cognitive closure' [1]. Leroy found that this effect is lessened when the prior tasks are completed under high time pressure. That is not to say workers should be placed under more pressure: a questionnaire (N=471) demonstrated a correlation between workplace stress and decreases in work performance [2]. Instead, related tasks could be ordered such that attention residue is minimised. Cai et al. undertook crowd-based studies that found that microtasks are completed faster when grouped with others on the same subject [3]. The effects of context switching extend throughout the organisational structure as well. In an empirical study, Edna and Santos found that both individuals in smaller teams and smaller teams as a whole are on average more productive [4]. They found a negative correlation between a team's productivity and their project size, complexity, and duration, and found that sharing workers in between teams harms productivity due to task switching. Both findings by Cai and Edna [3, 4] reinforce Leroy's claim that context switching leads to increased cognitive load [1].

An important source of context switching is interruptions. On the effectiveness of blocking digital interruptions, Mark et al. found that workers who were susceptible to online distractions felt more productive, however workers who already felt in control of their workload reported higher signs of stress since they used distractions to take breaks [5]. The approach Mark took is one that actively blocks distractions. An alternative would be a *passive* system such as the TimeToFocus tool employed by Borghouts et al. that informs the user of how much time it took them to recover from an interruption [6]. Both tools address self-interruptions however TimeToFocus only brings the effect of interruptions to attention, and was successful in reducing distractions. Although Borghouts did not measure stress, it may be that this lightweight approach would better suit a higher number of employees. Mark's approach ignores external interruptions as they may be necessary for work—for example, email. Batching external notifications may help to minimise context switching as the media often recommends for email¹. However Mark found in earlier research [7] that there is no correlation between email batching and higher productivity in most cases. This section has shown that the literature agrees that context switching hinders productivity but does not agree on one single approach for managing distractions and interruptions that is beneficial in all cases.

This section will explore some solutions designed to measure worker's propensity to be interrupted and improve productivity. Züger et al. developed FlowLight, a traffic-light representation of interruptibility [8]. Flowlight was a success—interruptions were reduced by 46% and 58% of participants claimed to feel more productive. However the measure was based on the participant's interaction with their computer which is not accurate as there exist high-intensity tasks such as debugging which don't require increased user input. Züger's earlier work showed that a machine learning model evaluating input from psycho-

¹6 *Email Principles To Boost Productivity And Reduce Stress*. URL: <https://productivityist.com/6-email-principles/> (visited on Sept. 3, 2020).

physiological sensors provides a better measure of interruptibility [9]. It was 78% accurate in the field study however it was significantly smaller than the FlowLight study (N=20 vs N=449). This discrepancy highlights the infeasibility of using such sensors in a real workplace. A flaw in both systems is that they may classify work breaks as interruptible. In this case, productivity may actually be reduced, as Epstein et al. found that an interrupted work break did not yield any reprieve for the worker [10].

Mandl et al. proposed and tested an instructive augmented reality overlay to help manual workers learn tasks faster [11]. The approach of guiding inexperienced operators through manufacturing processes was one of the case studies used to test the effectiveness of augmented reality in aiding productivity. The case studies found that for the simple manufacturing task of bending metal, the time taken to complete the process was reduced with AR compared to the control group, although no data was provided to back the claim. A further case study to investigate scalability showed many ‘task specific adjustments’ were required for more complex tasks. It was found overall that AR in its current state is a more disruptive solution than it is beneficial [11]. This section has shown that while using sensors and specific hardware can aid productivity, it is case dependent and would have adoption issues in a practical setting.

This section will discuss the limitations of the research above. One such limitation found across the research was that often the time period of the study was short. Most studies lasted around 2 weeks and multiple issues arise from this. Firstly, in studies which involved a significant change in worker routines [5, 6, 7, 8, 9, 10], the workers would likely need time to adapt and settle into these changes so behaviour in the first days of each study may be anomalous. Secondly, it’s possible that during the study a non-average work week occurred. This was specifically addressed in Epstein et al.(2016) in which a participant described the workweek as atypical [10]. Another limitation which spans multiple papers is the issue of survey accuracy and workers’ self-assessments. Züger et al.(2017) and Mark et al.(2018) mention that workers may not have logged all interruptions [8] or that the survey questions required participant interpretation [5]. This is much harder to analyse than more objective data gathered by the researchers themselves or by sensors as used in other parts of Züger et al.(2017). Further research should attempt to reduce subjectivity in survey questions and rely as little as possible on worker self-assessment, or aid the worker in self-assessment.

After analyzing the papers, it was found that breaks are briefly mentioned in papers that covered distractions and stress in the workplace. Many of these consider breaks to be distractions which should be avoided [2, 5, 6, 8] and only Epstein et al. [10] tackle the problem directly. To reduce the negative impact of attention residue on productivity, a transition phase in the form of a break could be introduced to reduce the distraction of a worker when they are switching tasks [1]. However the length or type of break remain unexplored, and the possibility that workers in larger groups need a larger or more personalized break schedule is not considered [4]. By analyzing what makes an effective and positive break we believe we may be able to design a product which can improve a worker’s focus and make a positive contribution to their work-related stress and productivity.

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